New features for GRNsight: a web application for visualizing models of small- to medium-scale gene regulatory networks

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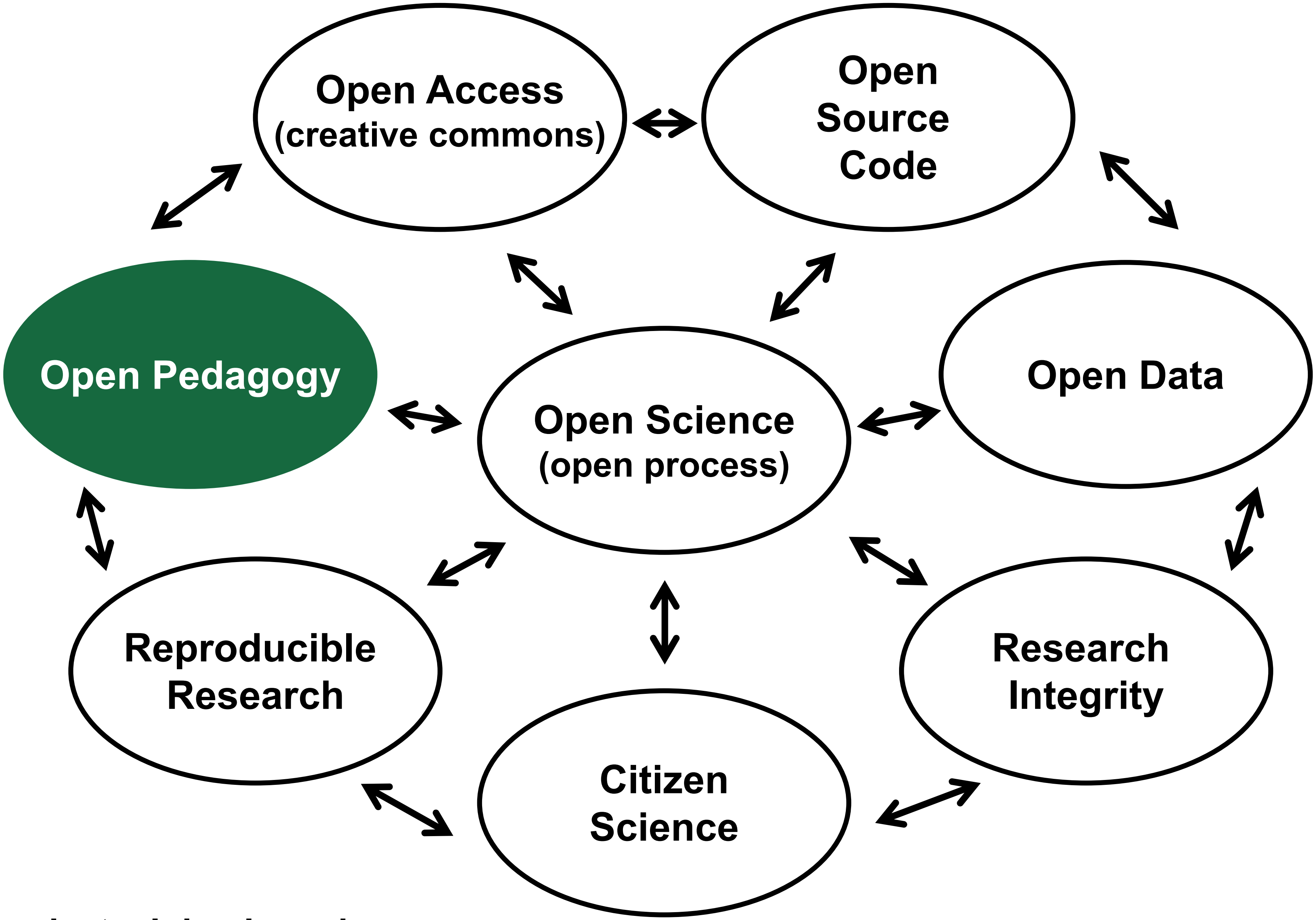
Bioinformatics Community Conference
Bioinformatics Open Source Conference
July 19-22, 2020

Loyola Marymount University
Frank R. Seaver
College of Science
and Engineering

Open Science Ecosystem

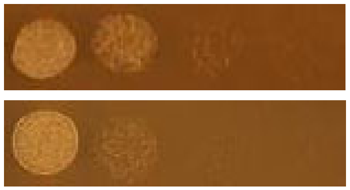


With thanks to John Jungck

GRNsight Is Developed and Used by Undergraduates in Courses and an Interdisciplinary Research Team

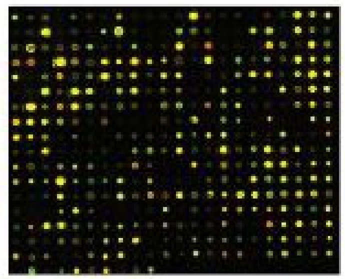
wt

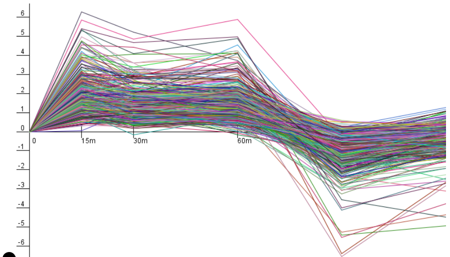
Δash1



15°C

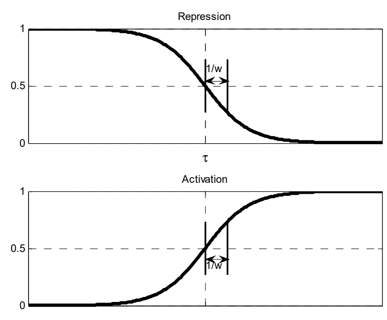
Cold shock microarray data from wt and TF deletion strains





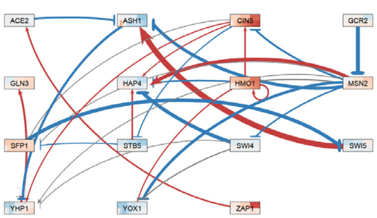
Normalization, statistical analysis, clustering

Derivation of gene regulatory networks from YEASTRACT



Dynamical systems modeling using GRNmap

Visualization of modeling results using GRNsight

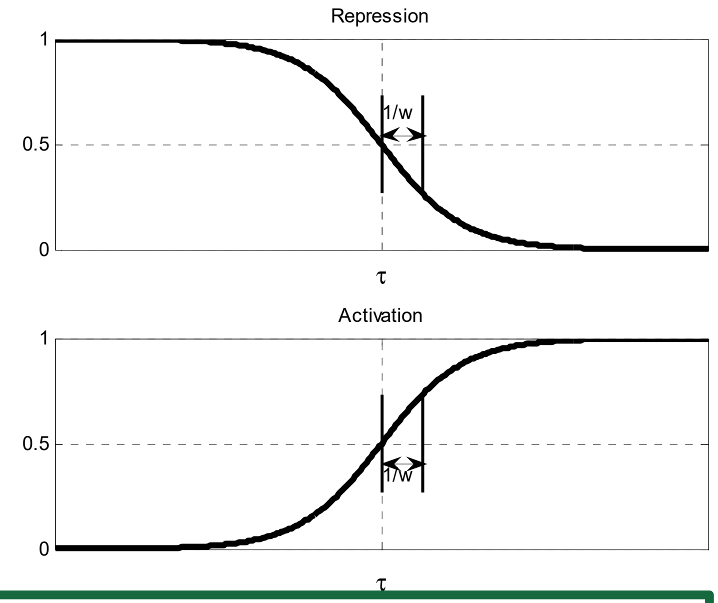


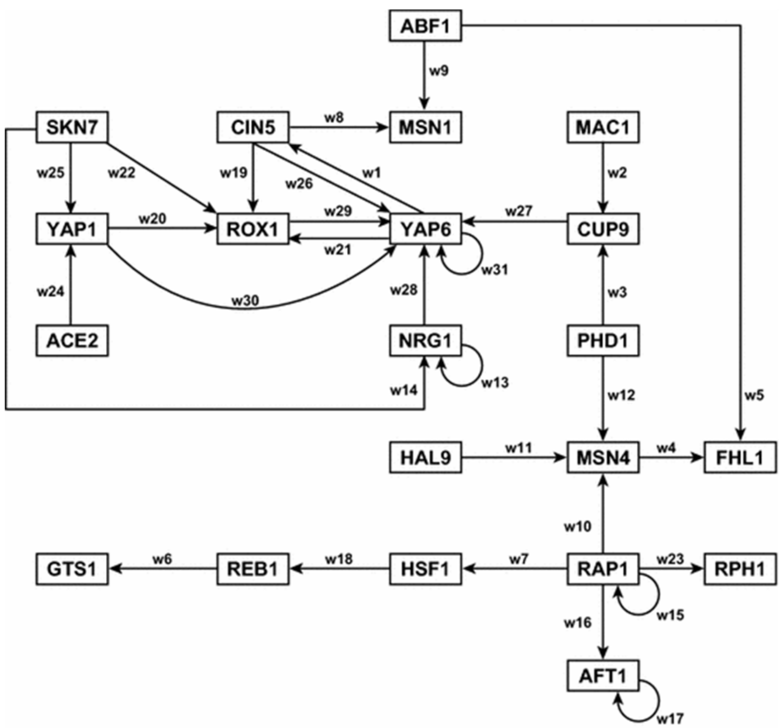
Interpretation, new questions, new experiments

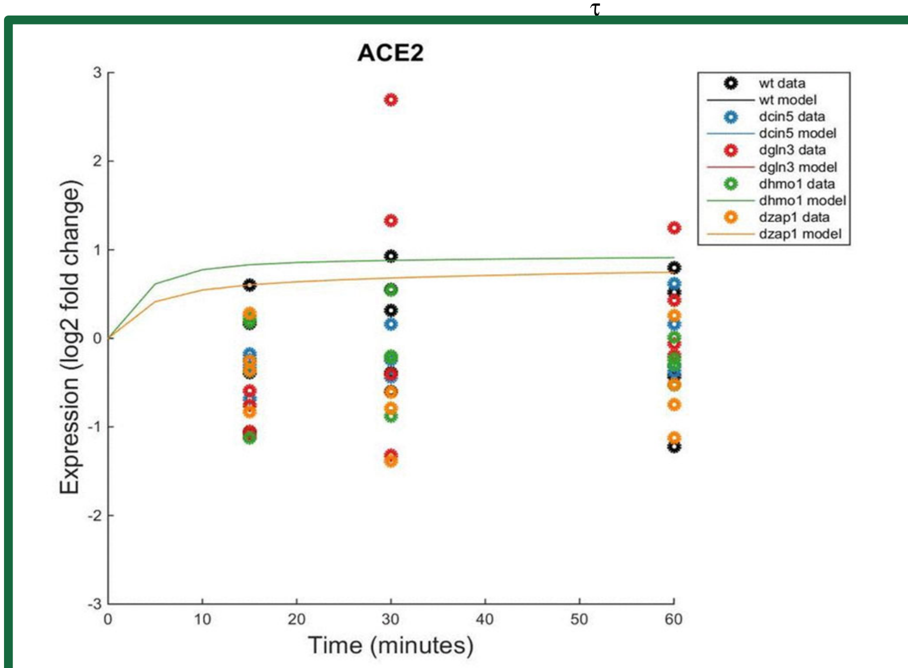
GRNmap: Gene Regulatory Network Modeling and Parameter Estimation

$$\frac{dx_i(t)}{dt} = \frac{P_i}{1+\exp\left(-\left(\sum_j \left(w_{ij}x_j(t)\right)-b_i\right)\right)} - d_i x_i(t)$$

Weight parameter, *w*, gives the direction (activation or repression) and magnitude of regulatory relationship.

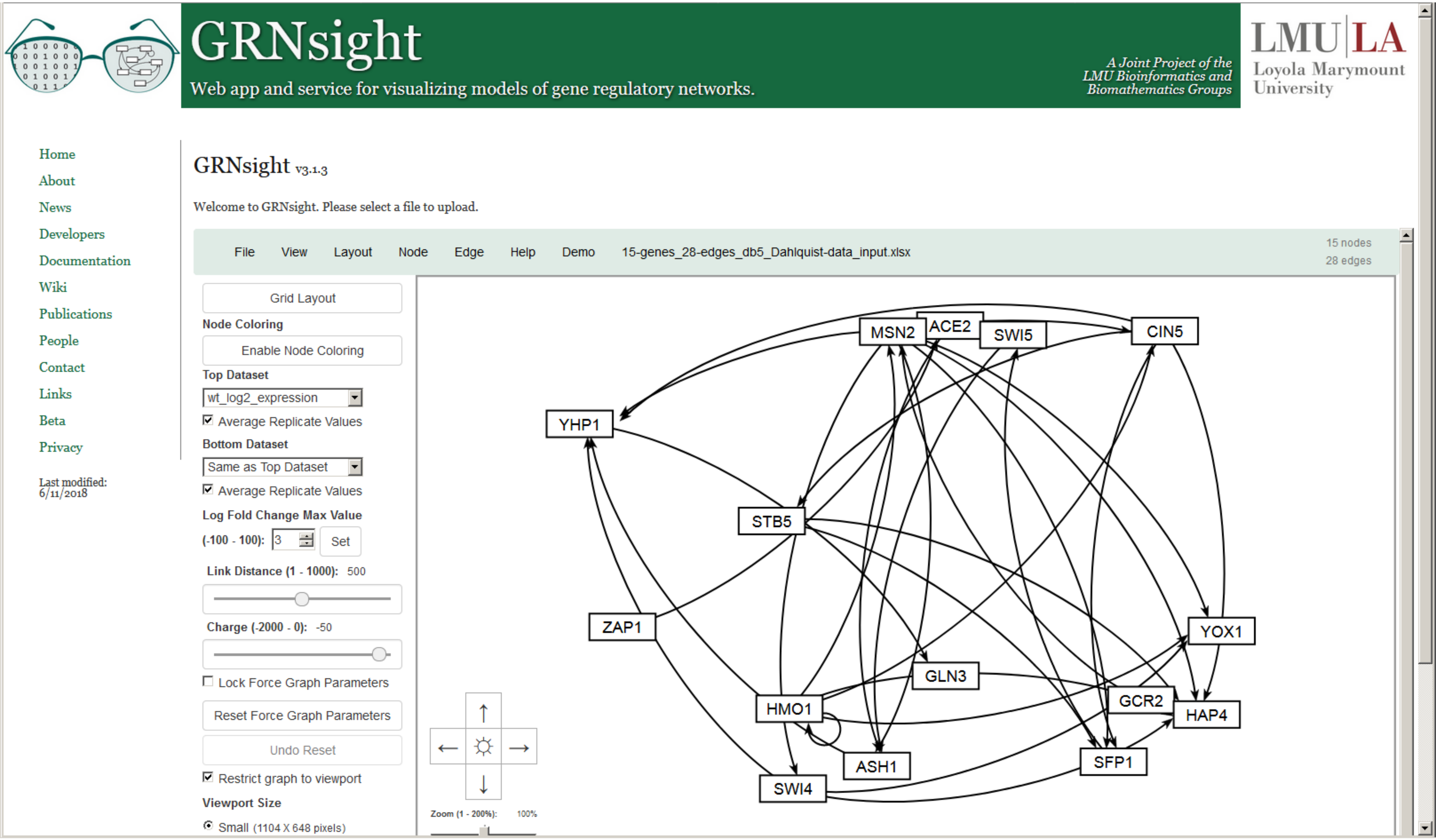






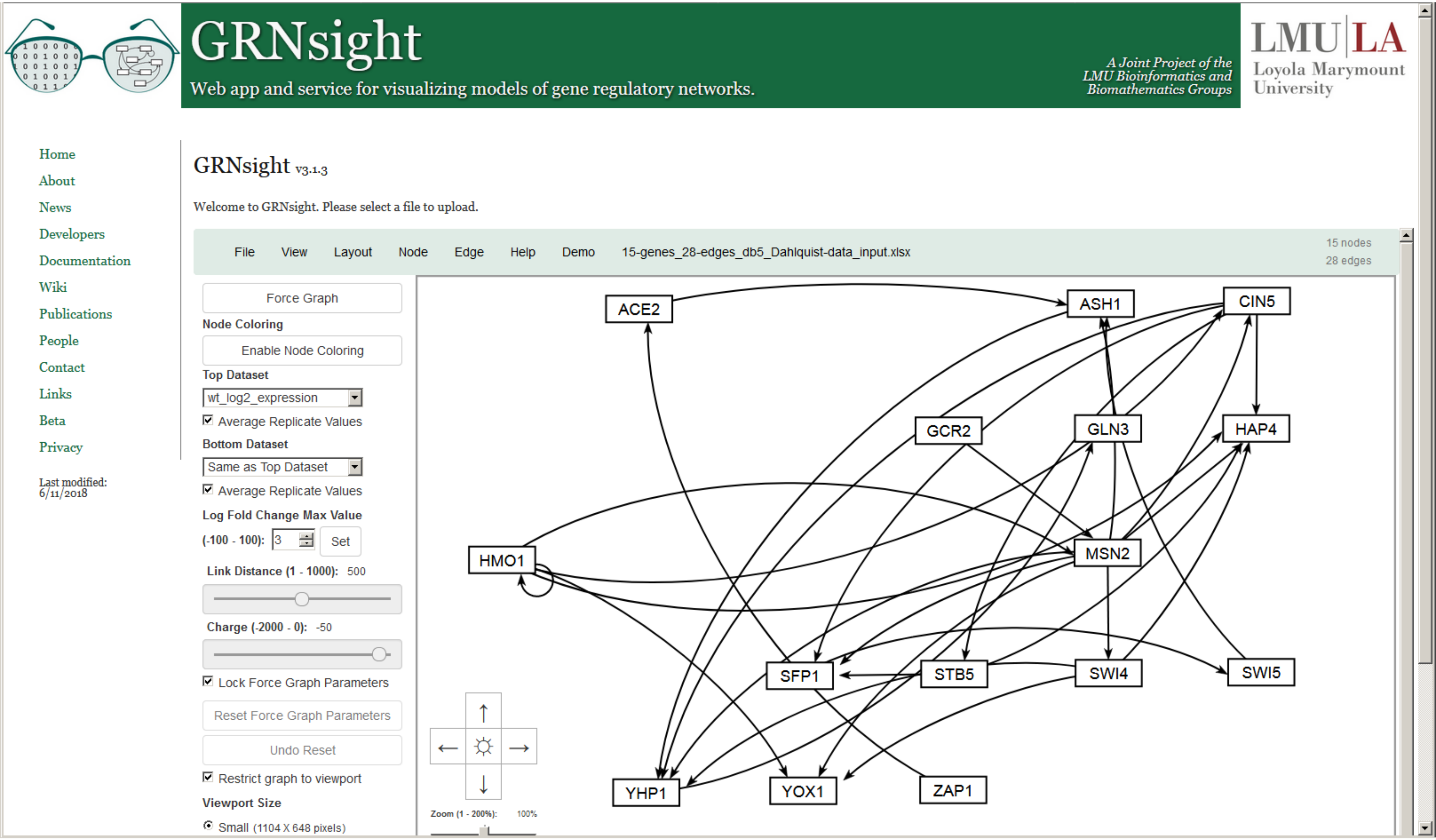
Dahlquist et al. (2015) *Bulletin of Mathematical Biology* 77: 1457.

GRNsight Rapidly Generates GRN Graphs Using Our Customizations to the Open Source D3 Library



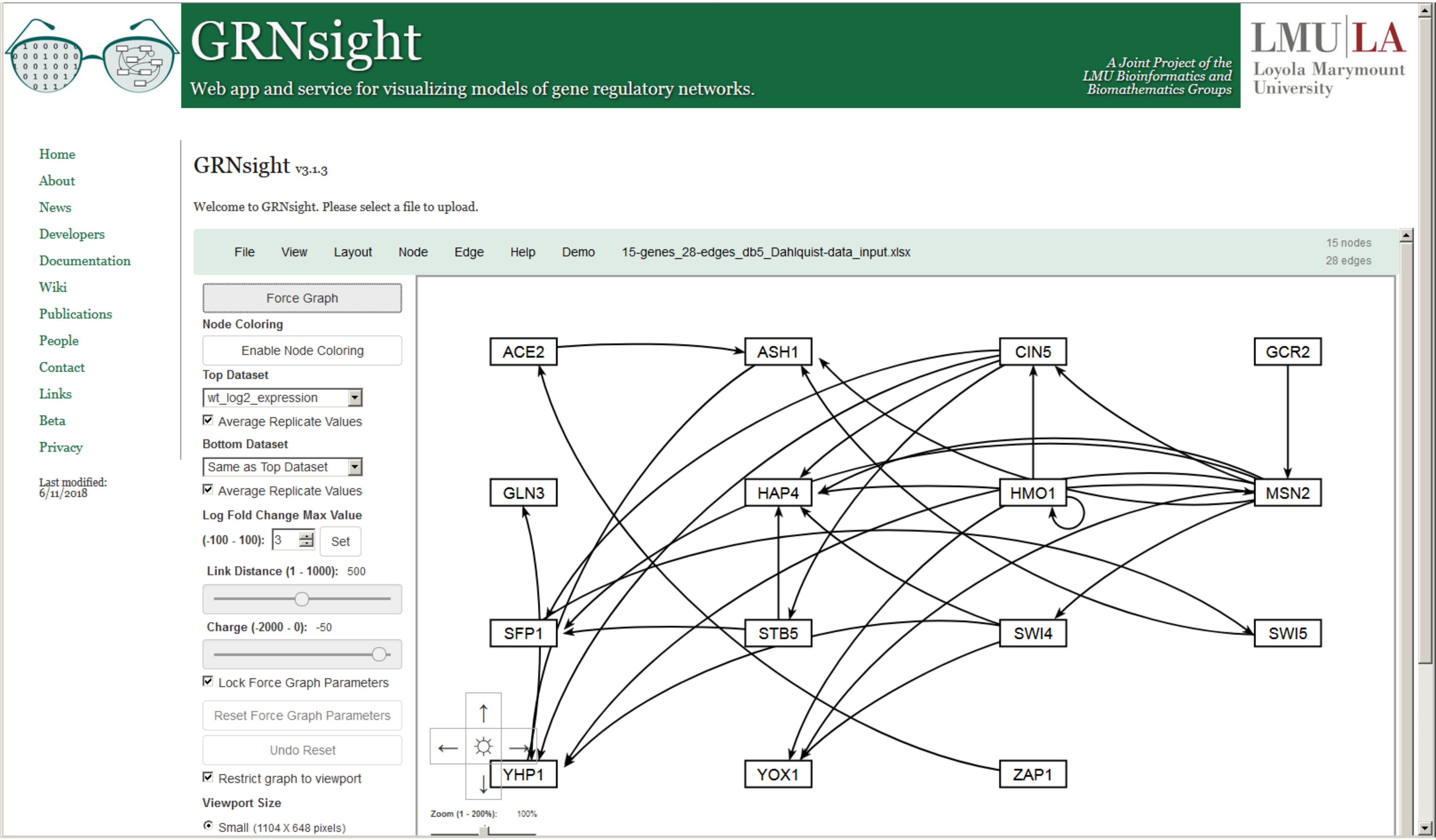
Dahlquist et al. (2016) *PeerJ Computer Science*; Version 1 available March 2014

Besides the Force Graph Layout, Users Can Rearrange the Nodes by Hand



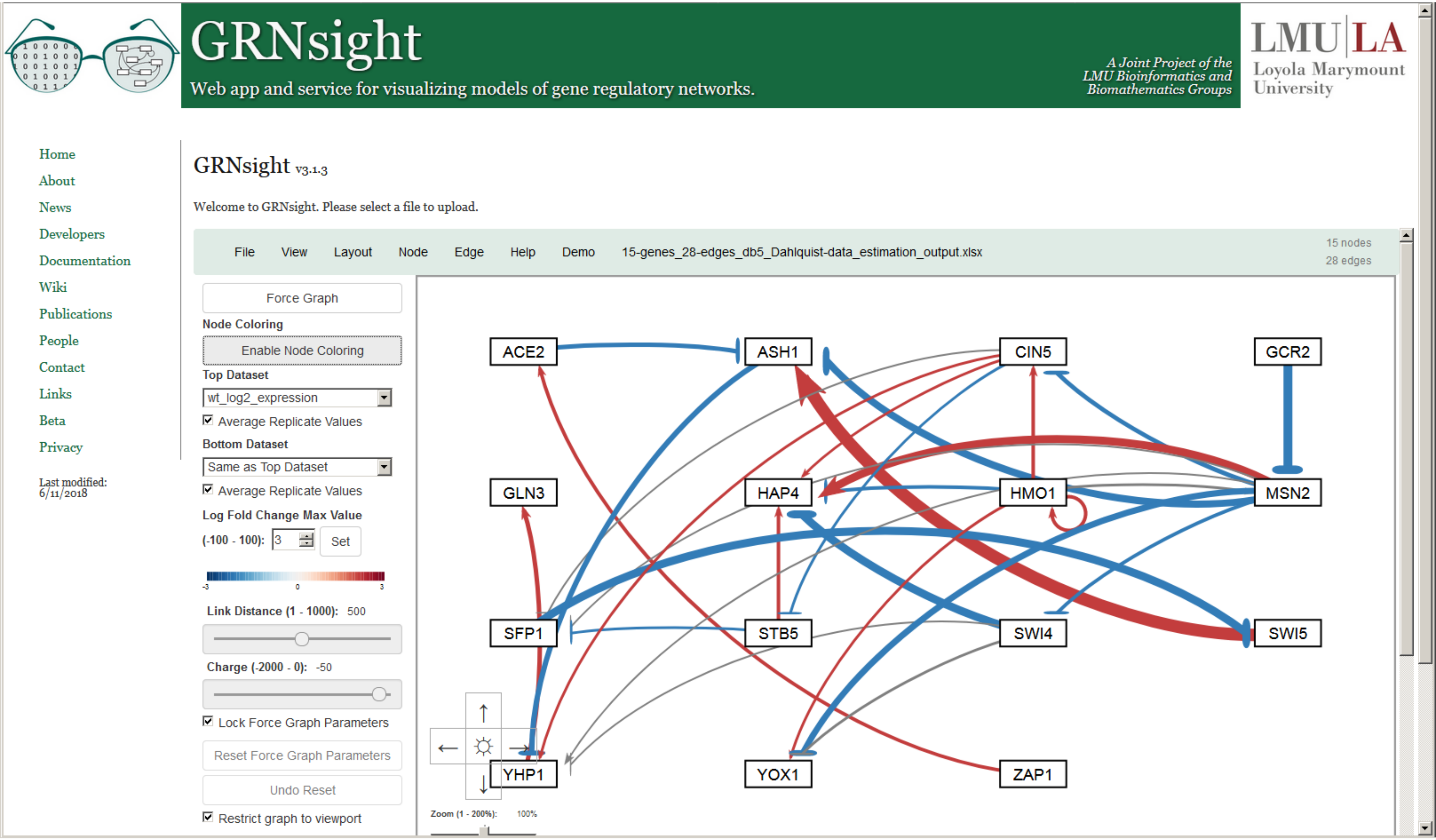
Dahlquist et al. (2016) *PeerJ Computer Science*

Or Choose the Convenient Block Layout Option



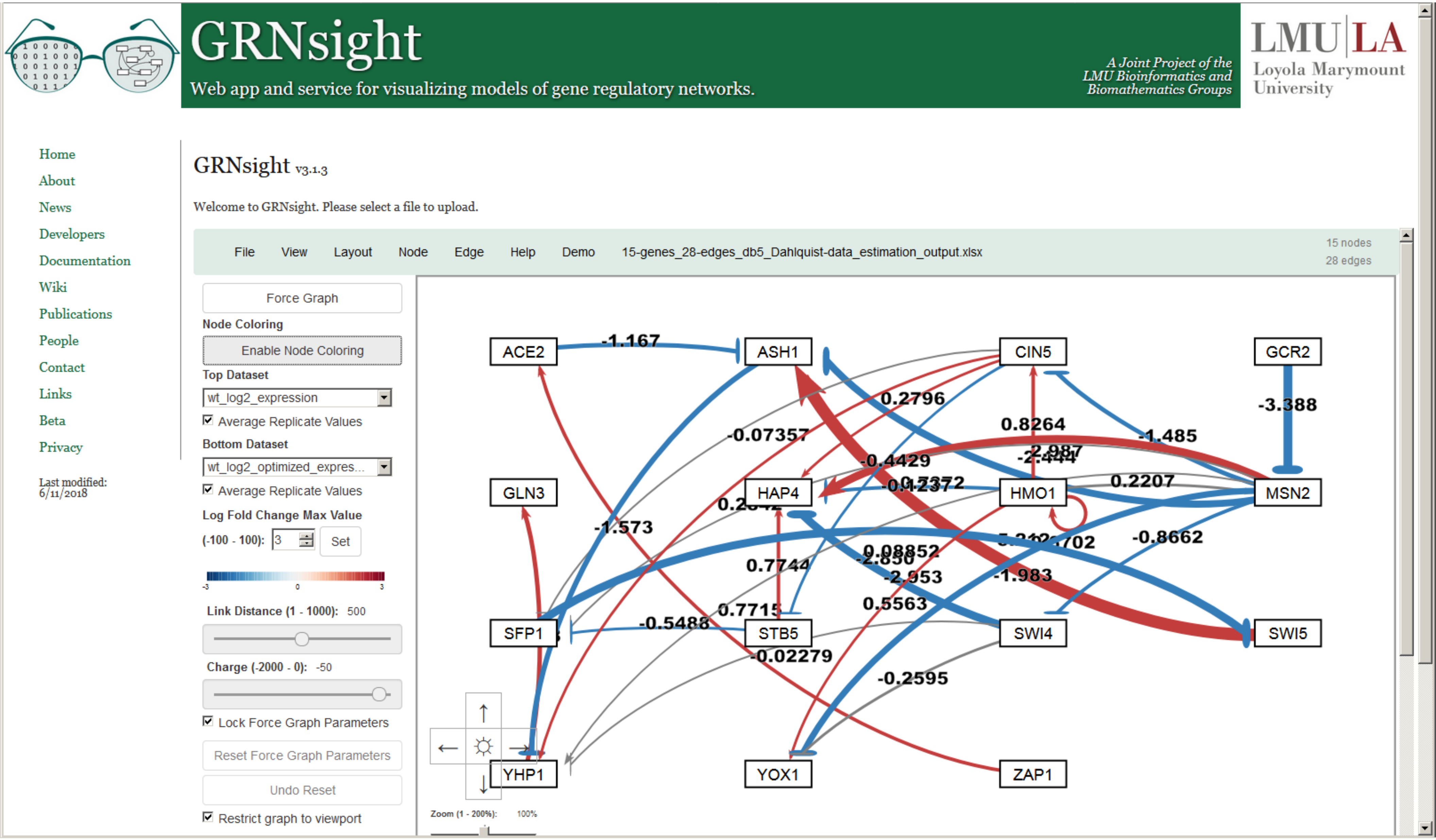
New in Version 3

When Model Results are Loaded, Edge Weight Values Are Indicated by Color, Thickness, and End Marker



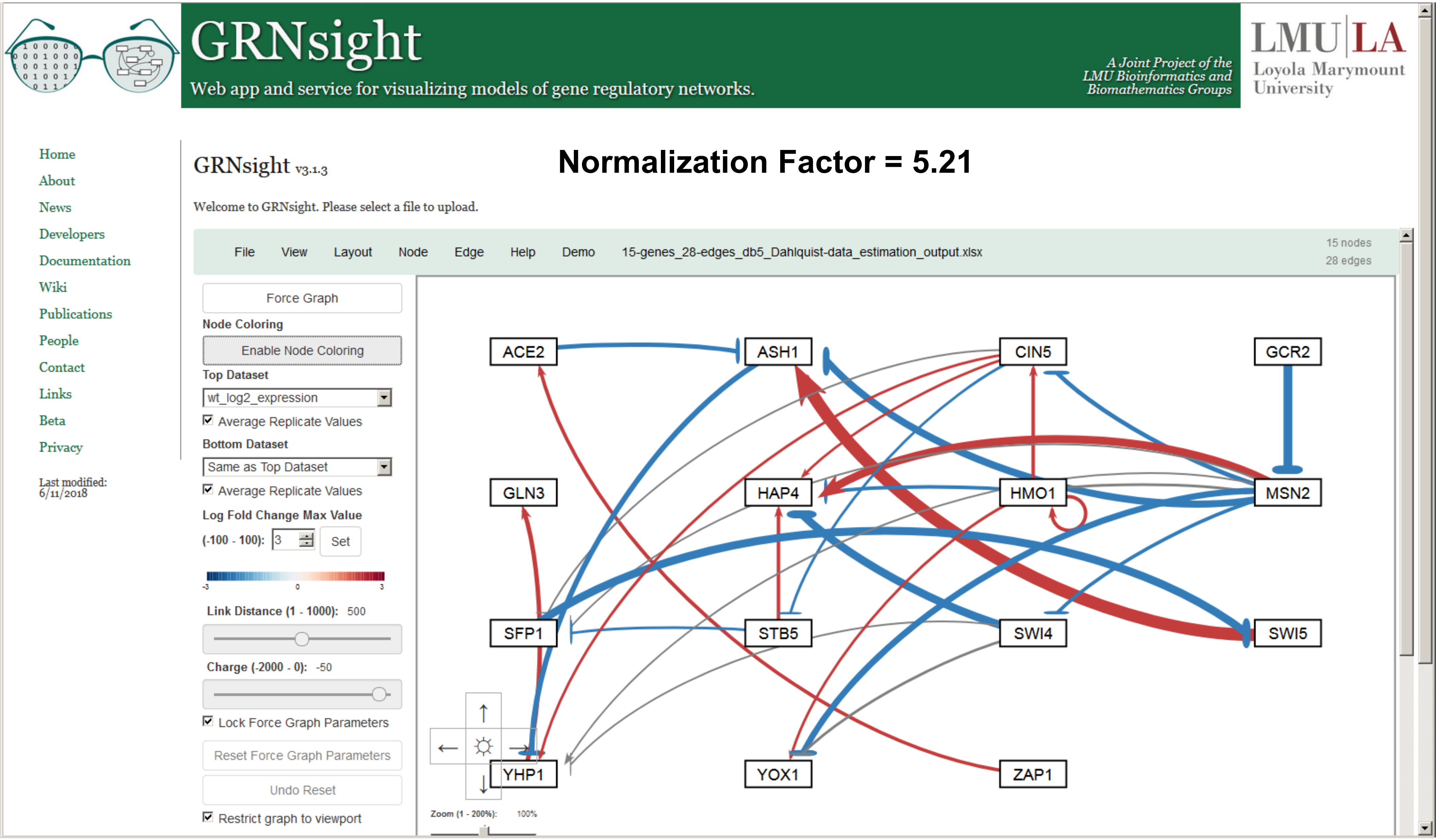
New color scheme in Version 3

Numerical Weight Values Can Be Displayed on the Graph



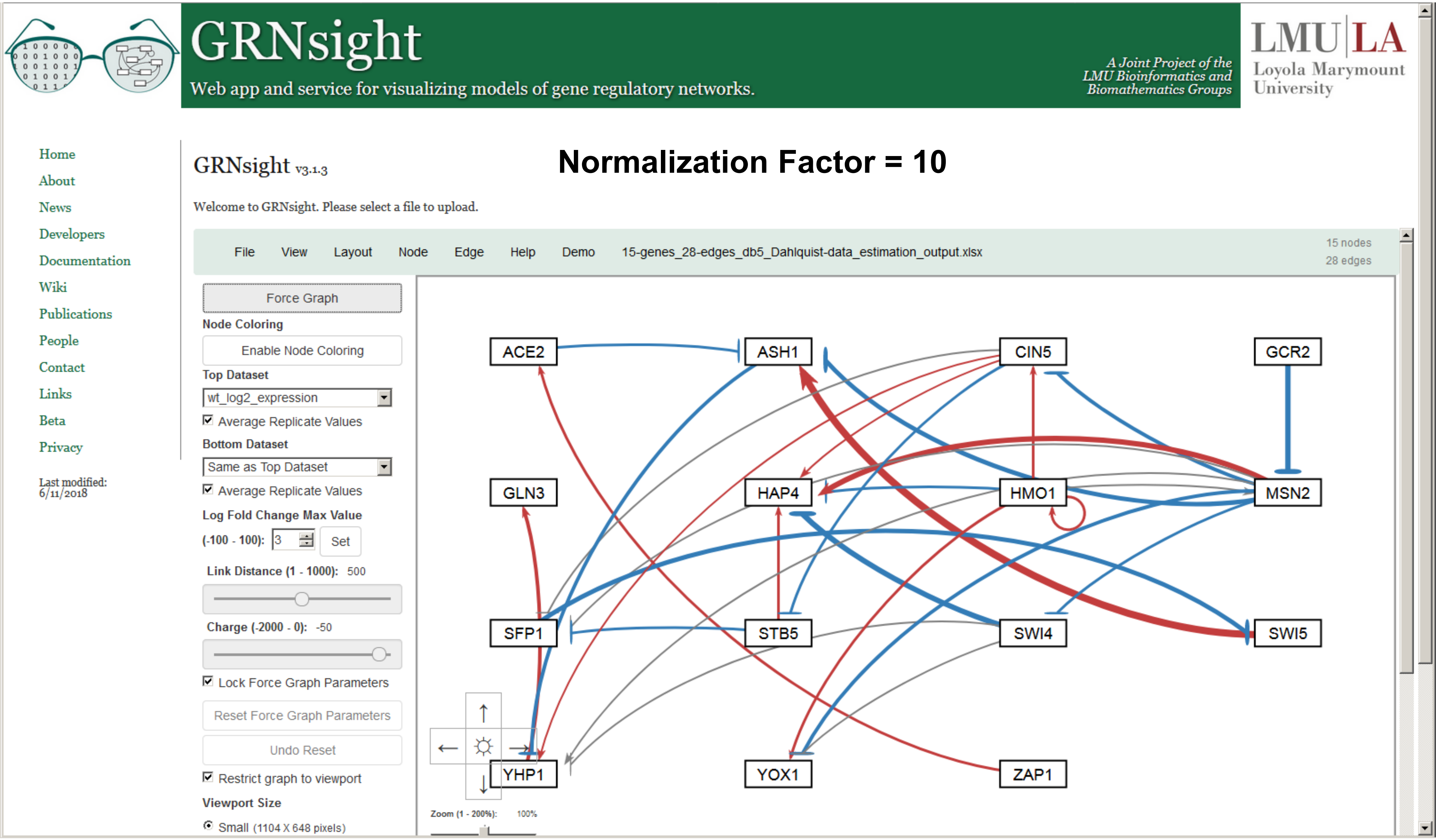
New in Version 2

Edge Thicknesses are Normalized to the Largest Magnitude Weight Value



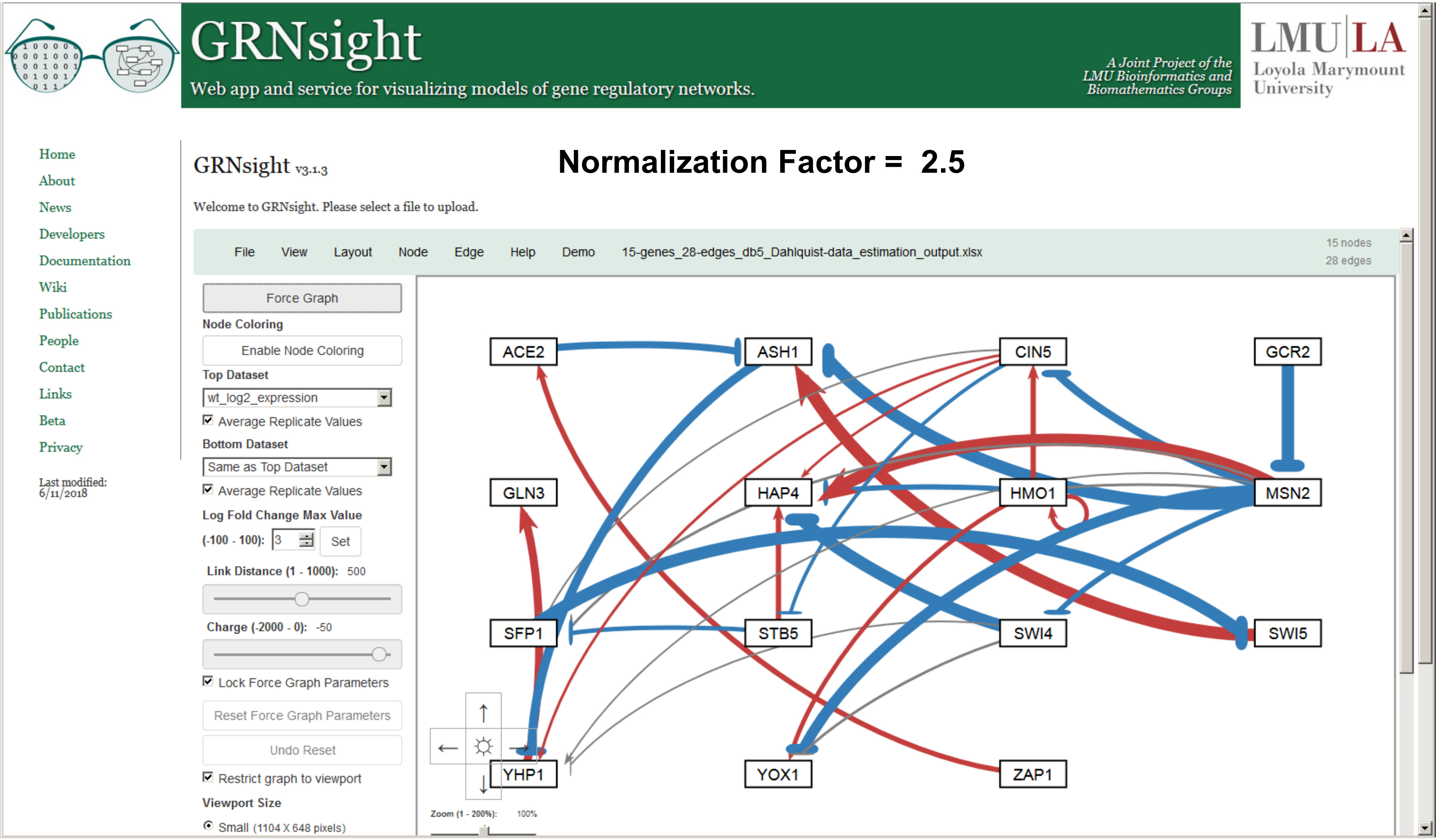
Version 1

And Can Be Modified to Allow Fair Visual Comparison between Different Graphs



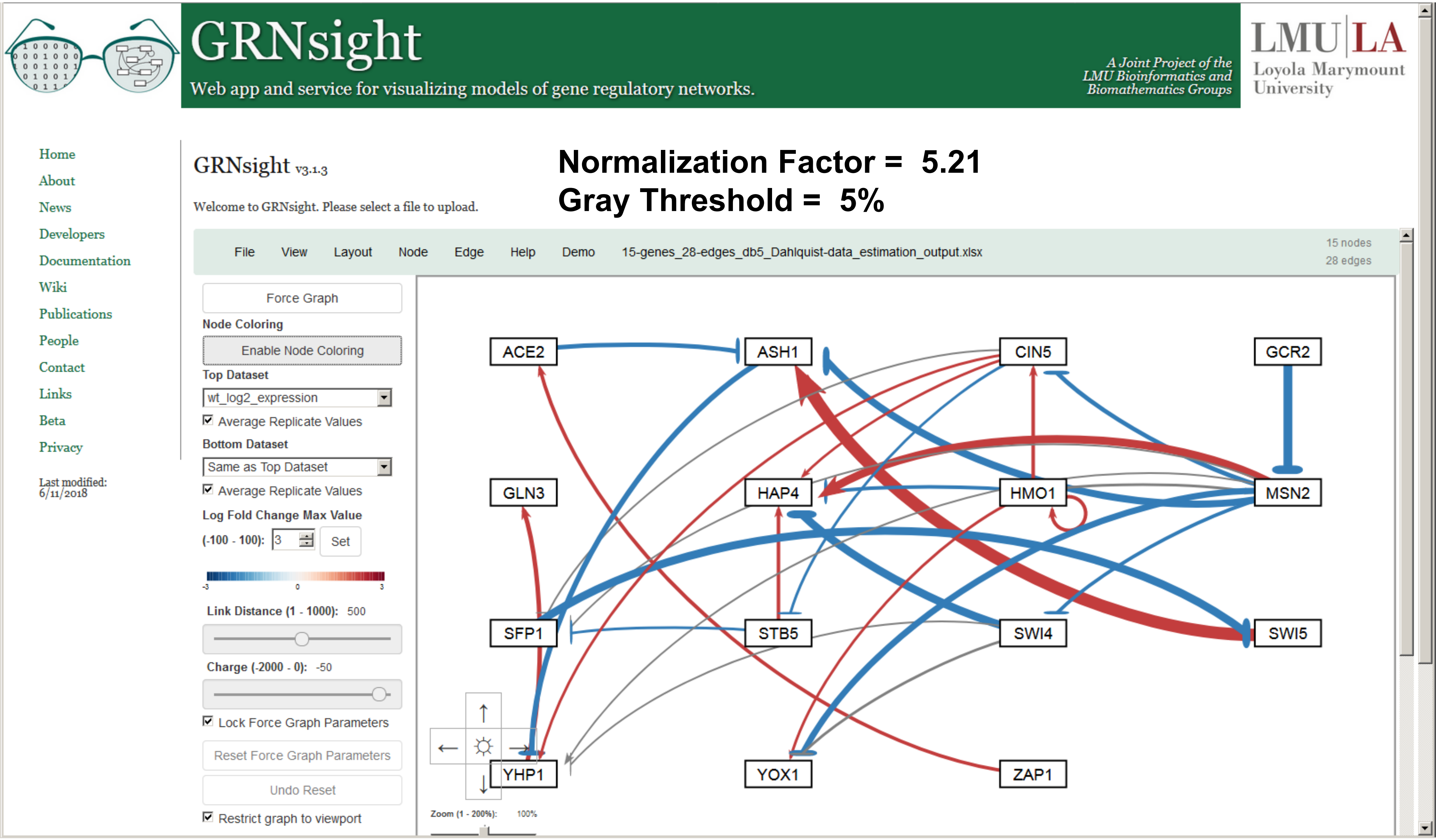
New in Version 2

And Can Be Modified to Allow Fair Visual Comparison between Different Graphs



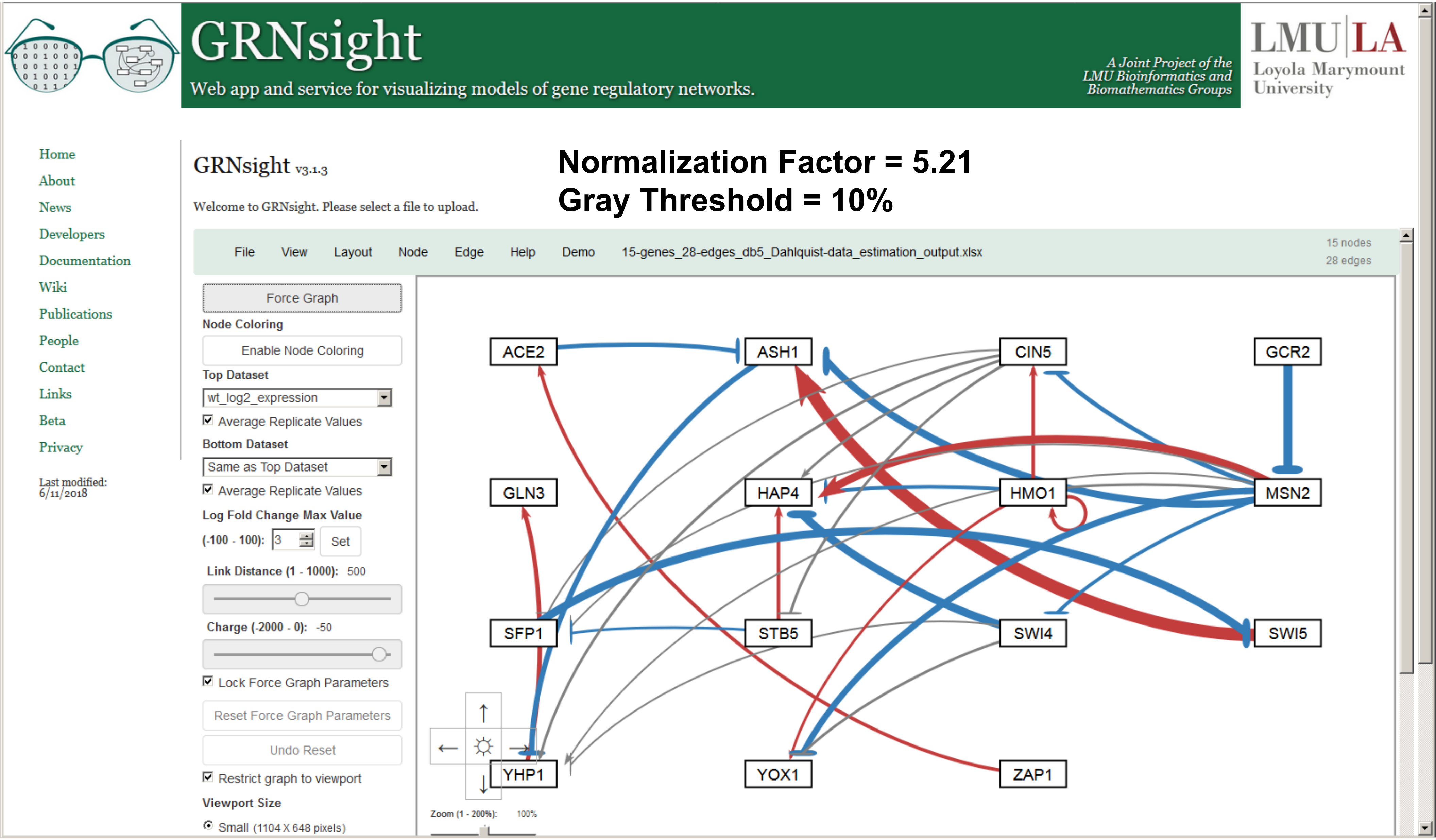
New in Version 2

Edges with Normalized Weight Values within 5% of Zero Are Displayed as Gray to Minimize Their Visual Strength



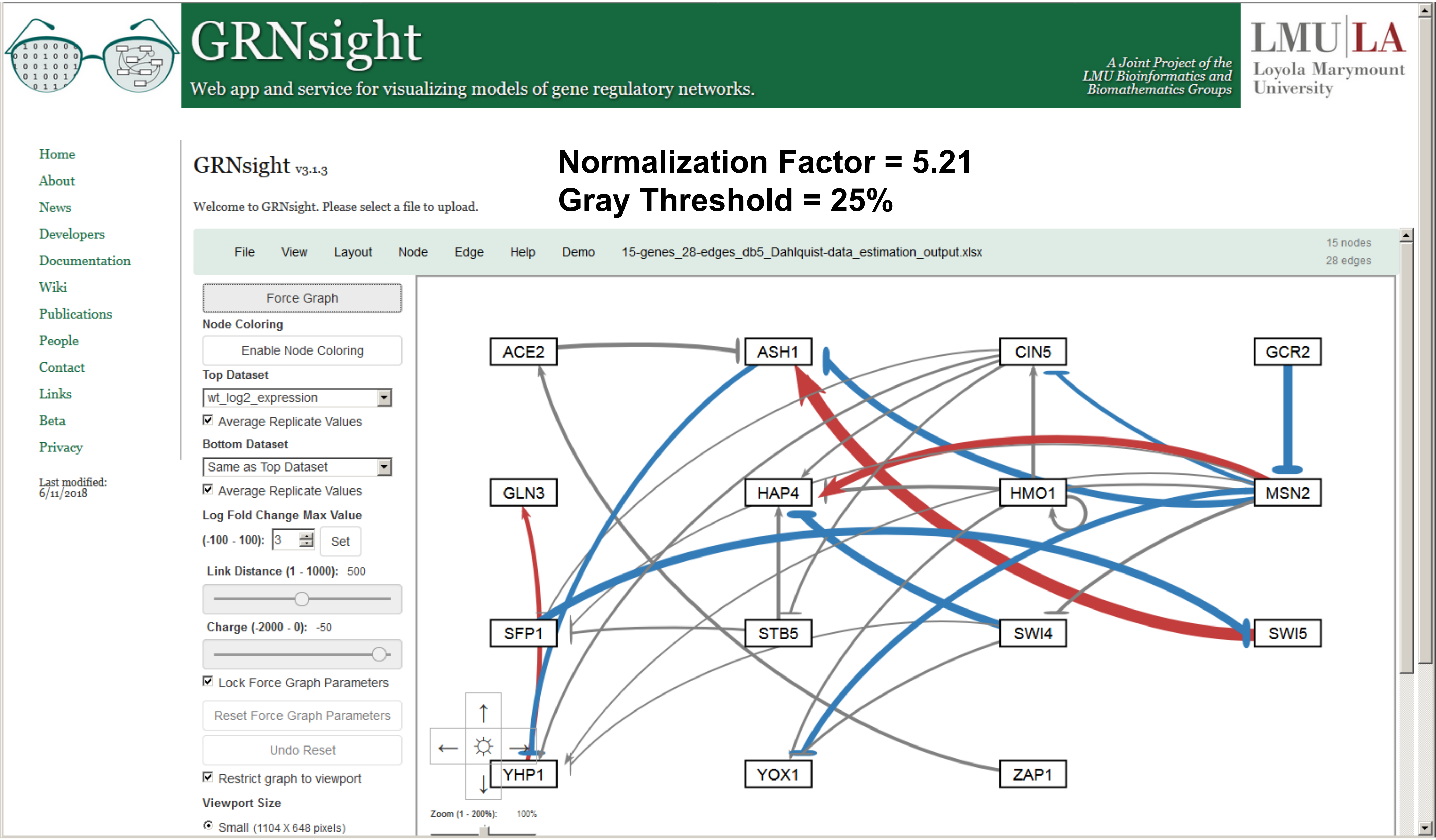
Version 1

Users Can Change the Gray Threshold with a Slider



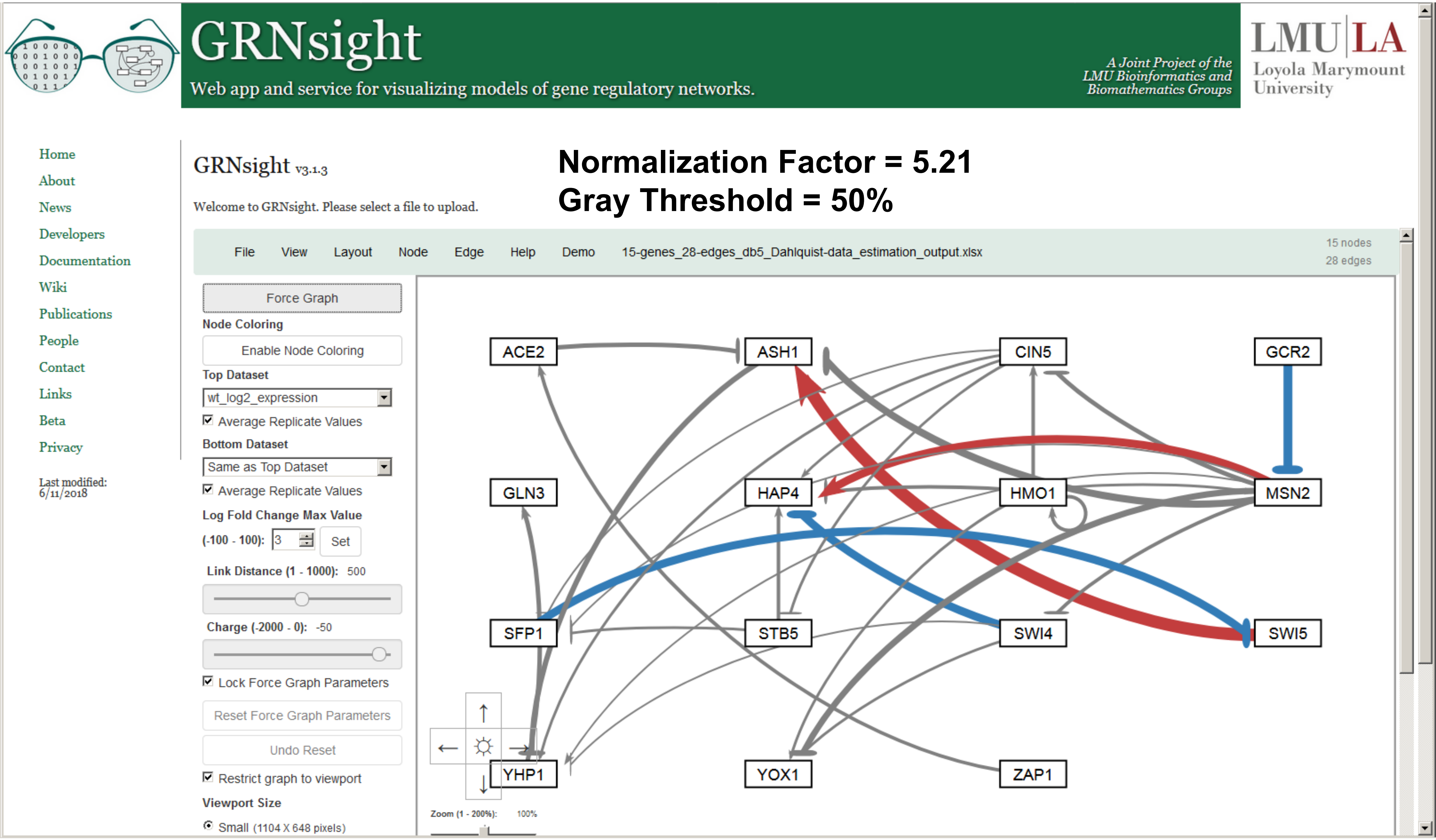
New in Version 2

Users Can Change the Gray Threshold with a Slider



New in Version 2

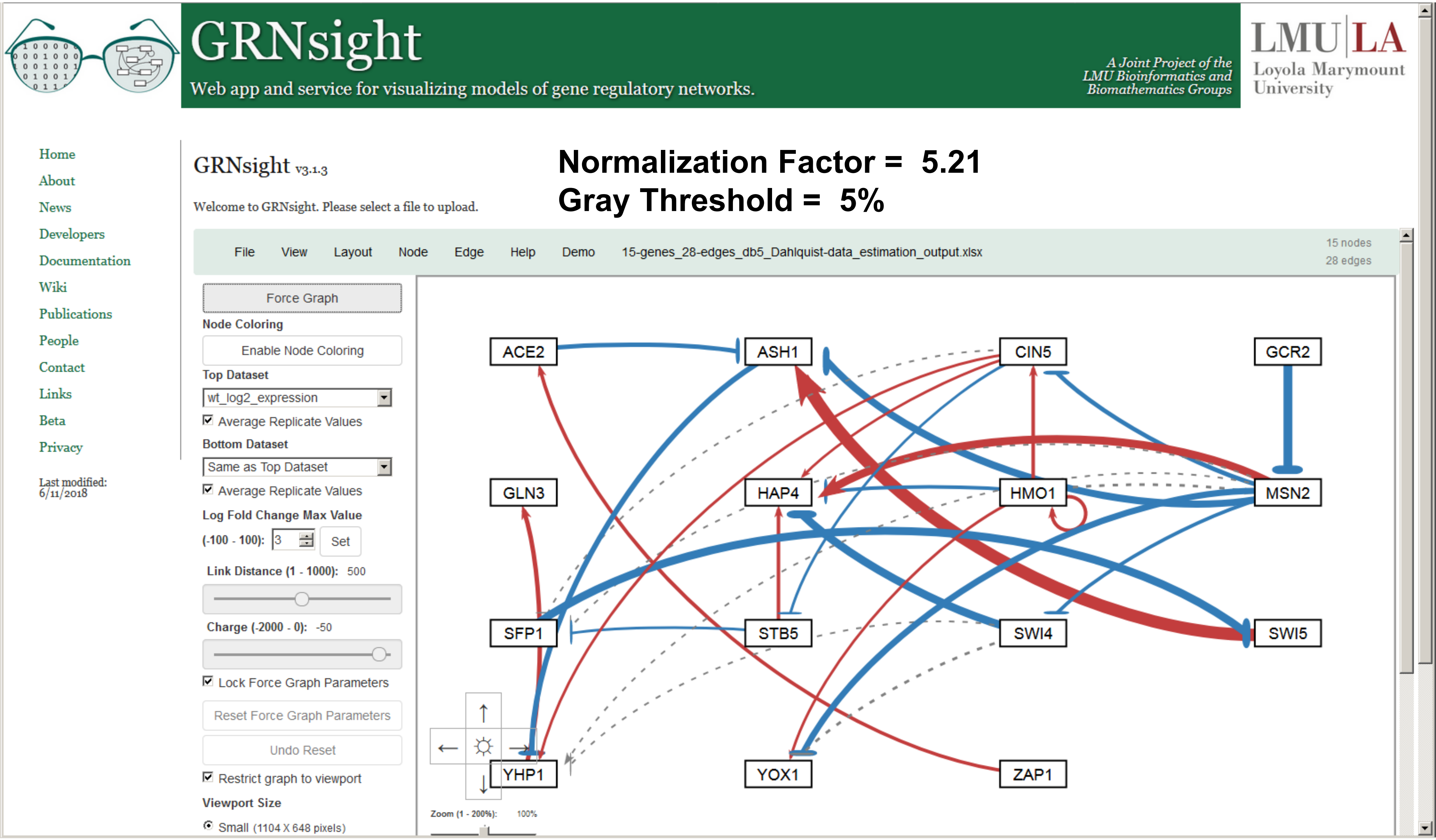
Users Can Change the Gray Threshold with a Slider



Normalization Factor = 5.21
Gray Threshold = 50%

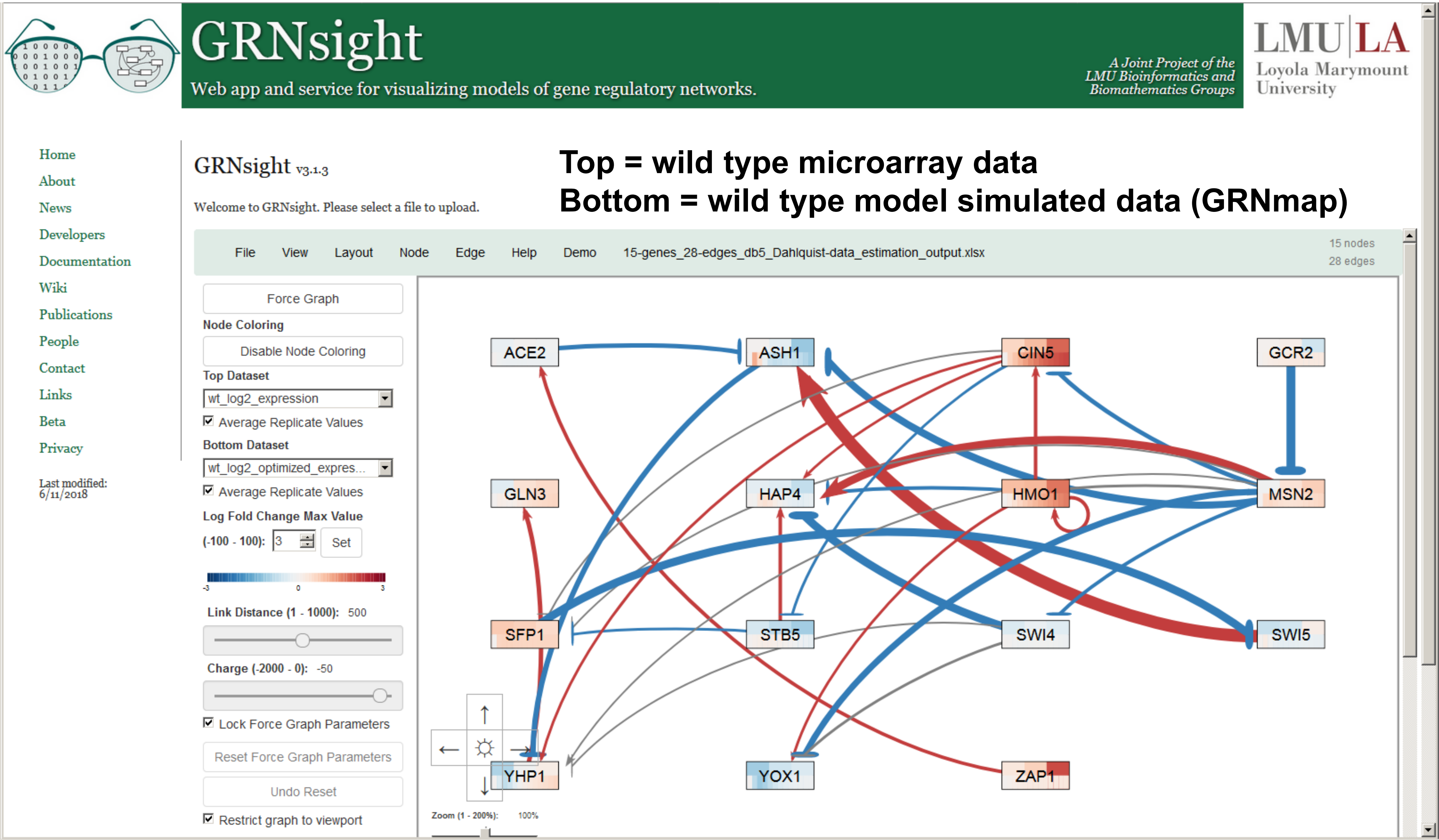
New in Version 2

Gray Edges Can Also Be Shown as Dashed Lines to Facilitate Visualization by Color Blind Users



New in Version 3

Nodes Can Display the Measured and/or Simulated Expression Data as Individual Heat Maps



Top = wild type microarray data
Bottom = wild type model simulated data (GRNmap)

Version 3

Right-clicking on a Gene Opens a New Web Page with Information Dynamically Retrieved from Source Databases

GRNsight

Web app and service for visualizing models of gene regulatory networks.

A Joint Project of the LMU Bioinformatics and Biomathematics Groups

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Last modified: 5/18/2018

GLN3

Saccharomyces cerevisiae (strain ATCC 204508 / S288c)

General Information | Protein Information | Regulation | Gene Ontology | Sources

General Information

SGD ID:	S000000842 [1]
NCBI Gene ID:	856763[4]
Ensembl ID:	Not found[3]
Uniprot ID:	GLN3_YEAST[2]
JASPAR ID:	MA0307.1[5]
Description:	Transcriptional activator of genes regulated by nitrogen catabolite repression; localization and activity regulated by quality of nitrogen source and Ure2p [1]
Species:	Saccharomyces cerevisiae (strain ATCC 204508 / S288c)[2]
Locus Tag:	YER040W[4]
JASPAR Family:	GATA-type zinc fingers[5]
JASPAR Class:	Other C4 zinc finger-type factors[5]
Chromosome Sequence:	NC_001137.3[4]

Protein Information

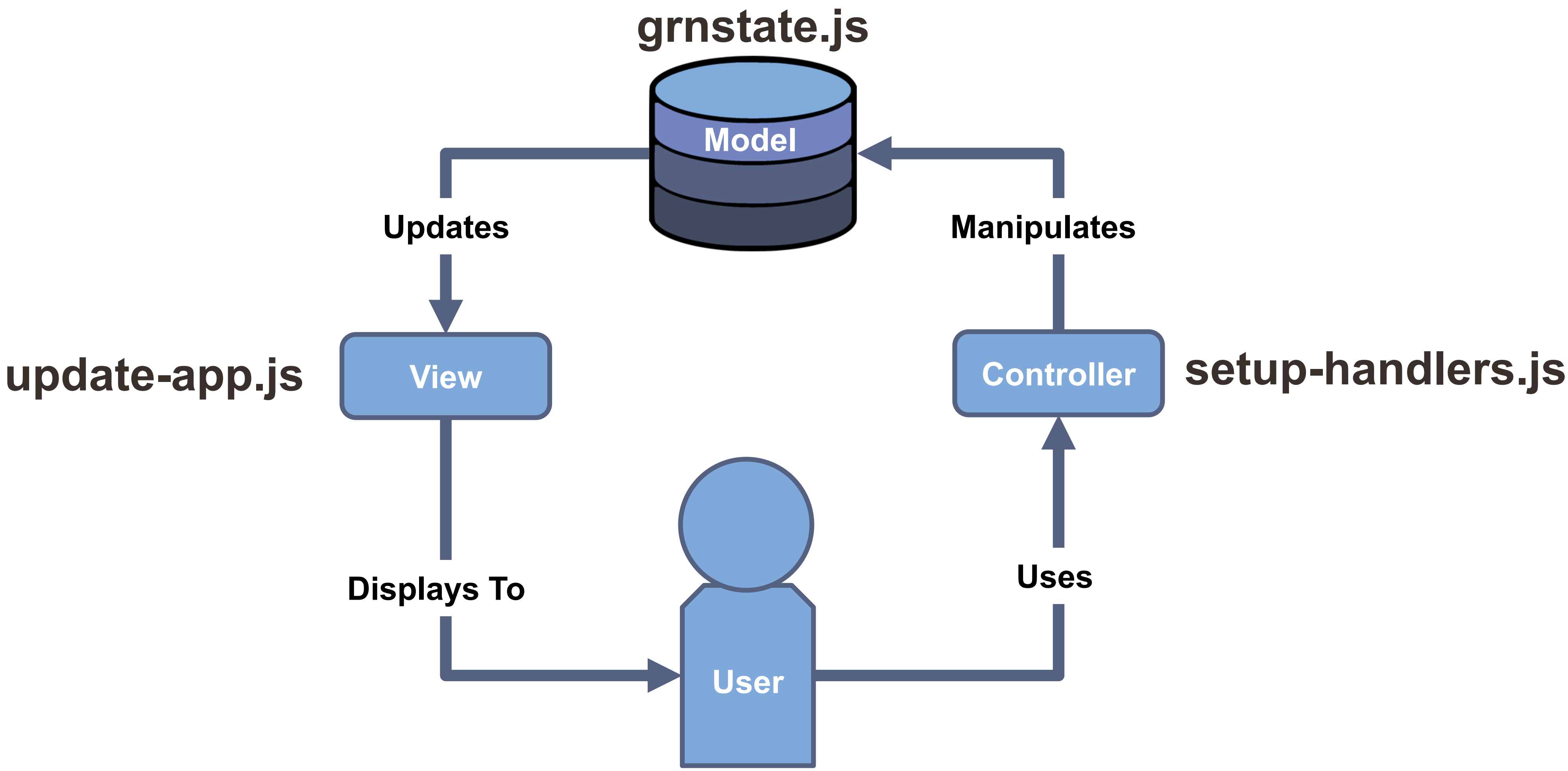
Regulation

Gene Ontology

Sources

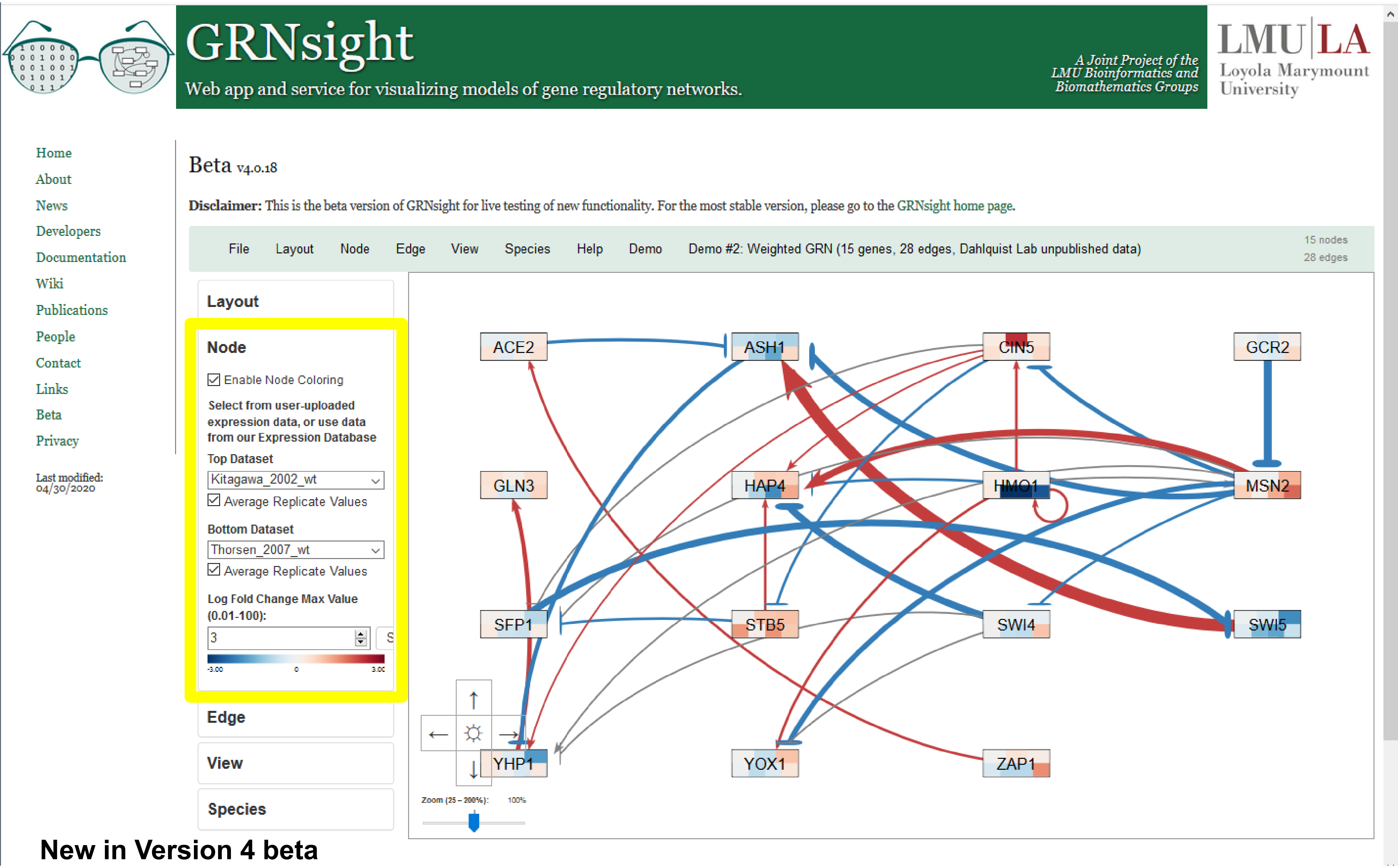
Version 3

Version 4 of GRNsight has been refactored to a Model-View-Controller Paradigm



12 files → 6 files

Data from a Backend PostgreSQL Expression Database Can Now Be Used to Color Nodes



New in Version 4 beta

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LMU Rains Research Assistant Program

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http://dondi.github.io/GRNsight/

https://github.com/dondi/GRNsight